6.4 Joint probability density function for continuous random variables

Although every pair of random variables (X, Y) has a joint distribution function $F_{X,Y}$, for discrete random variables it is usually more convenient to work with the joint probability mass function. Similarly, for continuous random variables it is often more convenient to use the joint probability density function, defined as follows.

Definition
We say that the random variables X and Y are **jointly continuous** random variables if there is a non-negative function $f_{X,Y}(x, y)$ with

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = 1$$

such that for any $a \leqslant b$ and $c \leqslant d$,

$$\boxed{\mathbb{P}(a \leqslant X \leqslant b,\ c \leqslant Y \leqslant d) = \int_c^d \int_a^b f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y}$$

and, more generally, for any 'nice' set $A \subseteq \mathbb{R}^2$,

$$\mathbb{P}\left((X,Y) \in A\right) = \iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y$$

The function $f_{X,Y}$ is called the **joint density function** of X and Y.

Remark: Note that

$$\int_c^d \left(\int_a^b f_{X,Y}(x,y)\ \mathrm{d}x\right)\mathrm{d}y = \int_a^b \left(\int_c^d f_{X,Y}(x,y)\ \mathrm{d}y\right)\mathrm{d}x$$

and more generally,

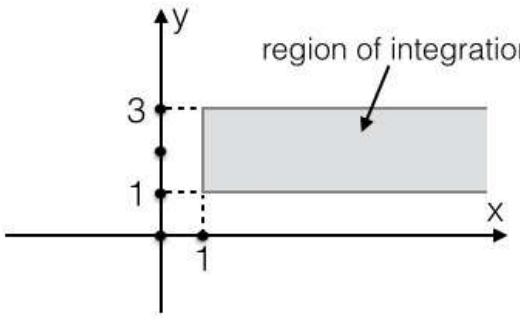
$$\iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = \iint_A f_{X,Y}(x,y)\ \mathrm{d}y\ \mathrm{d}x$$

That is, we can do the integrations either way round.

Example 6.10. Suppose that X and Y are jointly continuous with joint density function given by

$$f_{X,Y}(x,y) = \begin{cases} 3e^{-x-3y} & x, y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3)$
To compute this probability, we need to set up and evaluate a double integral of the joint density over the region in $\mathbb{R}^2$ that corresponds to the inequalities $1 \leq X$ and $1 \leq Y \leq 3$



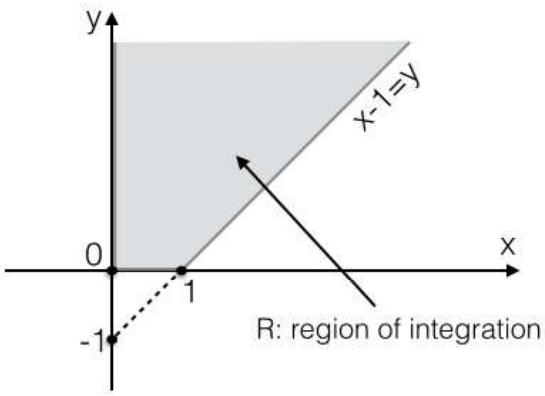
$$\begin{aligned}
\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3) &= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}x \right) \mathrm{d}y \\
&= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} 3e^{-x-3y}\ \mathrm{d}x \right) \mathrm{d}y \\
&= \int_{y=1}^{3} \left(\int_{x=1}^{\infty} e^{-x}\ \mathrm{d}x \right) 3e^{-3y} \mathrm{d}y \\
&= \int_{y=1}^{3} \left[-e^{-x} \right]_{x=1}^{\infty}\ 3e^{-3y} \mathrm{d}y \\
&= \int_{y=1}^{3} \left(e^{-1} - \lim_{x \to \infty} e^{-x} \right)\ 3e^{-3y} \mathrm{d}y \\
&= e^{-1} \int_{y=1}^{3} 3e^{-3y} \mathrm{d}y \\
&= e^{-1} \left[-e^{-3y} \right]_{y=1}^{3} \\
&= e^{-1}(e^{-3} - e^{-9}) \approx 0.0183
\end{aligned}$$

Alternatively (*exercise*), you can compute

$$\mathbb{P}(1 \leqslant X,\ 1 \leqslant Y \leqslant 3) = \int_{x=1}^{\infty} \left(\int_{y=1}^{3} f_{X,Y}(x,y)\ \mathrm{d}y \right) \mathrm{d}x$$

Find $\mathbb{P}(X - 1 \leqslant Y)$
This calculation is trickier because the region of integration for the double integral of the joint density is not a rectangle. This means that we have to think more carefully about the *order* of the integration in the double integral and the *limits* of integration in the double integral.

Step 1:
To work out how to set up the double integral, the first step is sketch the region in the plane where the inequality $X - 1 \leqslant Y$ is satisfied *and* $f_{X,Y}(x,y) > 0$ (we don't need to integrate over regions where $f_{X,Y}(x,y) = 0$).
This (infinite) region looks like this:



Step 2:
The next step is to set up and evaluate the double integral of the joint density over the region

of integration determined in Step 1 above.
Tip: Try to choose the order of integration to make the calculation as easy as possible.
So:

$$\begin{aligned}
\mathbb{P}(X-1 \leqslant Y) = \mathbb{P}\big((X,Y)\in R\big) &= \int_{y=0}^{\infty}\left(\int_{x=0}^{y+1} 3e^{-x}e^{-3y}\,\mathrm{d}x\right)\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left(\int_{x=0}^{y+1} e^{-x}\,\mathrm{d}x\right)3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left[-e^{-x}\right]_{x=0}^{y+1} 3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty}\left(1-e^{-(y+1)}\right)3e^{-3y}\,\mathrm{d}y \\
&= \int_{y=0}^{\infty} 3e^{-3y}-3e^{-(4y+1)}\,\mathrm{d}y \\
&= \left[-e^{-3y}+\frac{3}{4}e^{-(4y+1)}\right]_{y=0}^{\infty} \\
&= 1-\frac{3}{4}e^{-1}+\lim_{y\to\infty}\left(-e^{-3y}+\frac{3}{4}e^{-(4y+1)}\right) \\
&= 1-\frac{3}{4}e^{-1}\approx 0.724
\end{aligned}$$

Alternatively (*exercise*), you could integrate with respect to y first and then with respect to x, but to do the calculation in this way, you have to set up two double integrals, because of the shape of the region R (refer to sketch above):

$$\mathbb{P}(X-1\leqslant Y)=\int_{x=0}^{1}\left(\int_{y=0}^{\infty}3e^{-x}e^{-3y}\,\mathrm{d}y\right)\mathrm{d}x\;+\;\int_{x=1}^{\infty}\left(\int_{y=x-1}^{\infty}3e^{-x}e^{-3y}\,\mathrm{d}y\right)\mathrm{d}x=1-\frac{3}{4}e^{-1}$$

Properties of joint probability density functions
If $f:\mathbb{R}^2\to\mathbb{R}$ is the joint probability density function of a pair of random variables X,Y, then:

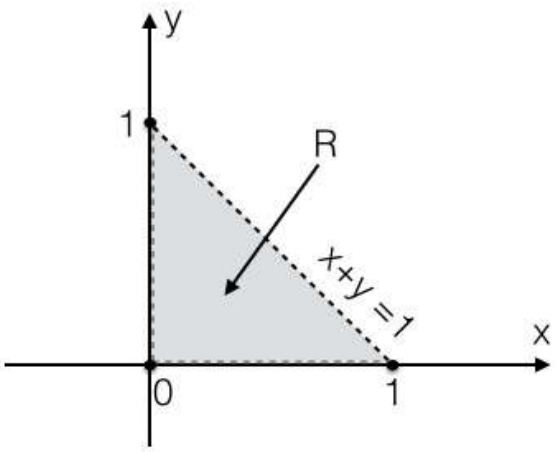
1. $f(x,y)\geq 0$ for all $(x,y)\in\mathbb{R}^2$
2. $\displaystyle\iint_{\mathbb{R}^2} f(x,y)\,\mathrm{d}x\,\mathrm{d}y=1$

Conversely, any function $f:\mathbb{R}^2\to\mathbb{R}$ satisfying the above two properties is a valid joint probability density function.

Example 6.11. Suppose that X and Y are jointly continuous random variables with joint density function given by

$$f_{X,Y}(x,y)=\begin{cases}60xy^2 & x>0,\ y>0,\ x+y<1\\ 0 & \text{otherwise}\end{cases}$$

Remark
The inequalities $x>0,\ y>0$ and $x+y<1$ describe the region of possible values for the ordered pair (X,Y), and this region is *not* a rectangle (due to the constraint $x+y<1$). It is important to understand what this region looks like in order to avoid making mistakes when you calculate probabilities. This region, R, is sketched below:



When working with joint density functions, **always sketch the region of non-zero density.**

To check that $f_{X,Y}(x,y)$ is a density, we note that

1. $f_{X,Y}(x,y) \geqslant 0 \quad \forall (x,y) \in \mathbb{R}^2$
2. The integral of $f_{X,Y}(x,y)$ over $\mathbb{R}^2$ is equal to 1:

$$\iint_{\mathbb{R}^2} f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y = \int_{y=0}^{1}\left(\int_{x=0}^{1-y} 60xy^2\,\mathrm{d}x\right)\mathrm{d}y \qquad \text{(Note the limits of integration!)}$$

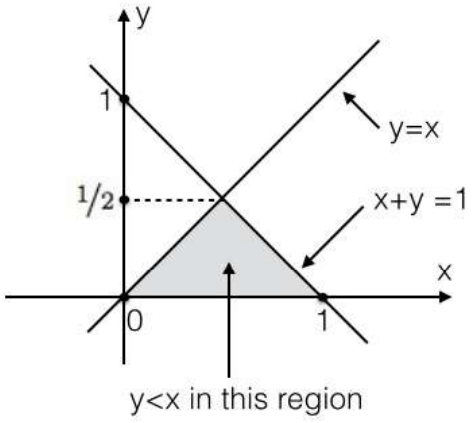
$$= \int_{y=0}^{1}\left[30x^2y^2\right]_{x=0}^{1-y}\,\mathrm{d}y = \int_{y=0}^{1} 30(1-y)^2y^2\,\mathrm{d}y$$

$$= 30\int_{y=0}^{1}(y^2-2y^3+y^4)\,\mathrm{d}y = 30\left[\frac{y^3}{3}-\frac{y^4}{2}+\frac{y^5}{5}\right]_{y=0}^{1} = 1$$

Determine $\mathbb{P}(Y < X)$

First step: To compute this probability, we have to integrate the joint density function $f_{X,Y}(x,y)$ over the region where $f_{X,Y}(x,y) > 0$ and $y < x$. So the first step is to sketch the region in the plane where these inequalities are satisfied in order to determine the region of integration. This sketch will help you to work out the correct limits of integration.

Here is a sketch of A, the region of integration:



Next step: The next step is to set up the double integral. To do this you will need to decide the order of integration. You should use the order of integration that makes the calculation as

uncomplicated as possible:

$$\mathbb{P}(Y < X) = \iint_A f_{X,Y}(x,y)\ \mathrm{d}x\ \mathrm{d}y = \int_{y=0}^{1/2} \left(\int_{x=y}^{1-y} 60xy^2\ \mathrm{d}x \right) \mathrm{d}y$$

Finally: Compute the double integral to obtain

$$\mathbb{P}(Y < X) = \frac{5}{16} \qquad \text{(Check this!)}$$

Remark: You can also compute $\mathbb{P}(Y < X)$ by integrating with respect to y first and then with respect to x. If you do the calculation this way, then you will need to set up *two* integrals:

$$\mathbb{P}(Y < X) = \int_{x=0}^{1/2} \left(\int_{y=0}^{x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x\ +\ \int_{x=1/2}^{1} \left(\int_{y=0}^{1-x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x$$

Example 6.12. Suppose that X and Y are jointly continuous random variables with joint density function $f_{X,Y}(x,y)$ given in example 6.11. Let $Z = X + Y$. Find the distribution function F_Z and density f_Z for the random variable Z.

Step 1: Determine ImZ
Since the joint density for X and Y is given by:

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

we must have $0 < X + Y < 1$. So $ImZ = (0,1)$.
Step 2: Determine F_Z
Since $ImZ = (0,1)$, we know that $F_Z(z) = 0$ for $z < 0$ and $F_Z(z) = 1$ for $z \geqslant 1$.
For $0 \leqslant z < 1$, we use the definition of the distribution function and Z to obtain:

$$F_Z(z) = \mathbb{P}(Z \leqslant z) = \mathbb{P}(X + Y \leqslant z)$$

To determine $\mathbb{P}(X + Y \leqslant z)$ we have to integrate the joint density $f(x,y)$ for X and Y over the region in the plane that corresponds to $0 < x + y \leqslant z$, i.e.

$$\begin{aligned}
\mathbb{P}(Z \leqslant z) = \mathbb{P}(X + Y \leqslant z) &= \int_{x=0}^{z} \left(\int_{y=0}^{z-x} f(x,y) \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{z} \left(\int_{y=0}^{z-x} 60xy^2 \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{z} \left[20xy^3 \right]_{y=0}^{z-x} \mathrm{d}x \\
&= \int_{x=0}^{z} 20x(z-x)^3 \mathrm{d}x \\
&= 20 \int_{x=0}^{z} xz^3 - 3x^2z^2 + 3x^3z - x^4\ \mathrm{d}x \\
&= 20 \left[\frac{x^2z^3}{2} - x^3z^2 + \frac{3x^4z}{4} - \frac{x^5}{5} \right]_{x=0}^{z} \\
&= 20 \left(\frac{z^5}{2} - z^5 + \frac{3z^5}{4} - \frac{z^5}{5} \right) \\
&= z^5
\end{aligned}$$

So the distribution function of Z is

$$F_Z(z) = \begin{cases} 0 & z < 0 \\ z^5 & 0 \leqslant z < 1 \\ 1 & z \geqslant 1 \end{cases}$$

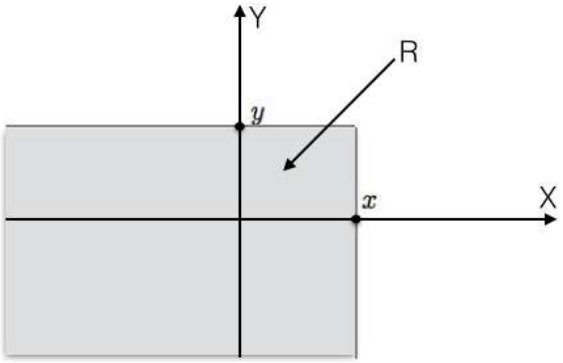
Step 3: Determine f_Z.
We differentiate F_Z to obtain $f_Z(z) = 5z^4$ for $0 < z < 1$ and $f_Z(z) = 0$ otherwise.

6.4.1 Relationship between joint distribution function and joint density function

By definition of the joint distribution function $F_{X,Y}(x, y)$ and the joint density $f_{X,Y}(x, y)$, we have for $x, y \in \mathbb{R}$

$$F_{X,Y}(x, y) \stackrel{def}{=} \mathbb{P}(X \leqslant x, Y \leqslant y)$$
$$F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u, v)\, \mathrm{d}u\, \mathrm{d}v = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, \mathrm{d}v\, \mathrm{d}u$$

where the region of integration corresponds to the (infinite) shaded region R below:



So, given the joint density $f_{X,Y}$, we can find the joint distribution function $F_{X,Y}$ by integrating the joint density $f_{X,Y}$ over an appropriate region.

Conversely, if $F_{X,Y}$ is *continuous* and is differentiable 'almost everywhere', then

$$f_{X,Y}(x, y) = \frac{\partial}{\partial x}\frac{\partial}{\partial y} F_{X,Y}(x, y) = \frac{\partial}{\partial y}\frac{\partial}{\partial x} F_{X,Y}(x, y)$$

wherever $F_{X,Y}$ is differentiable.

6.4.2 Marginal density functions

Suppose that X and Y are jointly continuous with joint density $f_{X,Y}$, then the **(marginal) density function** for X is given by

$$f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x, y)\, \mathrm{d}y \qquad \text{(integrate out } y\text{)}$$

and the **(marginal) density function** for Y is given by

$$f_Y(y) = \int_{x=-\infty}^{\infty} f_{X,Y}(x, y)\, \mathrm{d}x \qquad \text{(integrate out } x\text{)}$$

Proof. With $f_X(x)$ given by the above formula, for any $a, b \in \mathbb{R}$ with $a \leqslant b$ we have

$$\int_a^b f_X(x)\ \mathrm{d}x \overset{def}{=} \int_{x=a}^{b} \left(\int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \right)\ \mathrm{d}x = \mathbb{P}(a \leqslant X \leqslant b,\ -\infty < Y < \infty)$$
$$= \mathbb{P}(a \leqslant X \leqslant b) \qquad \text{(since there is no restriction on } Y\text{)}$$

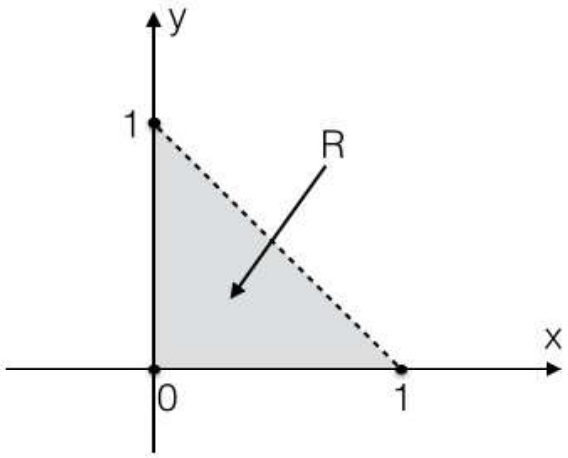
Furthermore, since the joint density satisfies $f_{X,Y}(x,y) \geqslant 0$ for all $x, y \in \mathbb{R}$, it follows that $f_X(x) = \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \geqslant 0$.
So the function $f_X(x)$ as defined above satisfies the properties of the density function for X.
A similar argument shows that the function $f_Y(y)$ as defined above is the density function for the random variable Y.

Example 6.11 continued. Suppose that X and Y are jointly continuous random variables with joint density function given by

$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Determine the marginal density f_X for X.

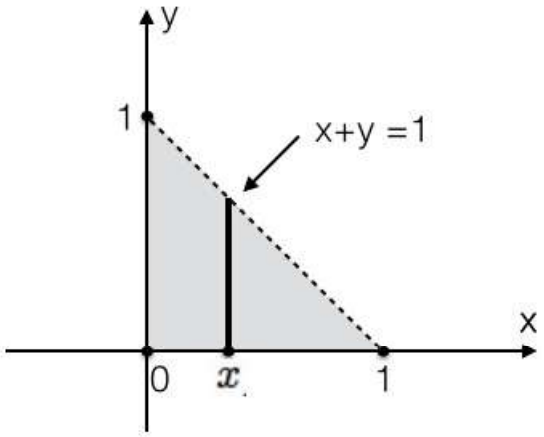
First step: Look at the region R where the joint density $f_{X,Y}(x,y)$ is non-zero.
We can determine from this region $\mathrm{Im}X$, the set of possible values for X.



From this picture it is clear that $\mathrm{Im}X = (0,1)$ since points where $f_{X,Y}(x,y) > 0$ occur for $0 < x < 1$. So we have $f_X(x) = 0$ if $x \leqslant 0$ or $x \geqslant 1$.

For $0 < x < 1$,

$$\begin{aligned} f_X(x) &= \int_{y=-\infty}^{\infty} f_{X,Y}(x,y)\ \mathrm{d}y \\ &= \int_{y=0}^{1-x} f_{X,Y}(x,y)\ \mathrm{d}y \qquad \text{(The limits of integration are based on the diagram below)} \\ &= \int_{y=0}^{1-x} 60xy^2\ \mathrm{d}y \\ &= \left[20xy^3\right]_{y=0}^{1-x} = 20x(1-x)^3 \\ &= 20x - 60x^2 + 60x^3 - 20x^4 \end{aligned}$$



Remark: You can check (*exercise*) that this is a valid density:

$$\int_{-\infty}^{\infty} f_X(x)\ \mathrm{d}x = \int_0^1 (20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x = 1.$$

Exercise: Use f_X to show that $\mathbb{P}(\frac{1}{4} \leqslant X \leqslant \frac{1}{2}) = \frac{57}{128}$.

Expected value and variance of X

Once we have computed $f_X(x)$, the marginal density for X, we can use this to compute $\mathbb{E}(X)$ and $\mathbb{V}\mathrm{ar}(X)$:

$$\begin{aligned} \mathbb{E}(X) = \int x f_X(x)\ \mathrm{d}x &= \int_0^1 x(20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x \\ &= \int_0^1 20x^2 - 60x^3 + 60x^4 - 20x^5\ \mathrm{d}x \\ &= \left[\frac{20x^2}{3} - 15x^4 + 12x^5 - \frac{10x^6}{3}\right]_0^1 \\ &= \frac{20}{3} - 15 + 12 - \frac{10}{3} = \frac{1}{3} \end{aligned}$$

Exercise: Show $\mathbb{V}\mathrm{ar}(X) = \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2 = \frac{2}{63}$

Exercise: Show that the marginal density f_Y for Y is given by

$$f_Y(y) = \begin{cases} 30y^2 - 60y^3 + 30y^4 & 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Use $f_Y(y)$

1. to compute $\mathbb{P}(\frac{1}{4} \leqslant Y \leqslant \frac{1}{2})$, and

2. to show that $\mathbb{E}(Y) = \frac{1}{2}$ and $\mathbb{V}\text{ar}(Y) = \frac{1}{28}$

Remark
The example above illustrates the general principle that given the joint density $f_{X,Y}(x, y)$ for the variables X and Y, we can always find the marginal density functions f_X and f_Y for the individual variables X and Y. Just as in the discrete case, the converse is *not* true. Given f_X and f_Y, it is not possible to determine the joint density $f_{X,Y}(x, y)$ for X and Y.

Independent jointly-continuous random variables
Recall that X and Y are independent if and only if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y) \qquad \forall (x, y) \in \mathbb{R}^2$$

So if X and Y are jointly continuous random variables with joint density function $f_{X,Y}(x, y)$, then, by differentiating the equation above, we see that X and Y are *independent* if and only if

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \qquad \forall (x, y) \in \mathbb{R}^2$$

i.e. joint density function = product of the marginal density functions **if independent**.

More generally, the random variables $X_1, X_2, \ldots, X_n$ with joint density function $f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n)$ are *independent* if and only if

$$f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n) = f_{X_1}(x_1)f_{X_2}(x_2) \ldots f_{X_n}(x_n) \qquad \forall (x_1, \ldots x_n)$$

Remarks

1. If we know that continuous random variables X and Y are independent, then we can find their joint density function from their marginal probability density functions using the relationship $f_{X,Y}(x, y) = f_X(x)f_Y(y)$. Can only do this if we know that X, Y are independent.
2. When the variables $X_1, X_2, \ldots, X_n$ depend on *different experiments* which are *physically independent*, then it is reasonable to assume that $X_1, X_2, \ldots, X_n$ are (statistically) independent and we can compute their joint density function by multipying together their marginal density functions:

$$f_{X_1 \ldots X_n}(x_1, x_2, \ldots, x_n) = \underbrace{f_{X_1}}_{\substack{\text{density} \\ \text{for } X_1}}(x_1) \cdot \underbrace{f_{X_2}}_{\substack{\text{density} \\ \text{for } X_2}}(x_2) \cdots \underbrace{f_{X_n}}_{\substack{\text{density} \\ \text{for } X_n}}(x_n)$$

3. In general, knowing the marginal density functions $f_{X_1}, f_{X_2}, \ldots, f_{X_n}$ is not sufficient to be able to find the joint probability density function $f_{X_1, X_2, \ldots, X_n}$
4. *Given* the joint density $f_{X,Y}$ for X and Y (say), we can compute f_X and f_Y and check whether the condition for independence is satisfied.

- If $f_{X,Y}(x,y) = f_X(x)f_Y(y) \quad \forall (x,y) \in \mathbb{R}^2$ then $X,\ Y$ are independent
- If $f_{X,Y}(x,y) \neq f_X(x)f_Y(y)$ for some $(x,y) \in \mathbb{R}^2$ then $X,\ Y$ are not independent

Example 6.11 continued. In Example 6.11, we were given the joint density function
$f_{X,Y}(x,y) = 60xy^2$ for $x > 0,\ y > 0,\ x + y < 1$
and have seen that the marginal density functions corresponding to this are
$f_X(x) = 20x - 60x^2 + 60x^3 - 20x^4$ for $0 < x < 1$
$f_Y(y) = 30y^2 - 60y^3 + 30y^4$ for $0 < y < 1$
Clearly $f_{X,Y}(x,y) \neq f_X(x)f_Y(y)$, so the random variables X and Y are not independent. (In this example, you can check this simply by looking at the regions of non-zero density; for any point (x,y) such that $0 < x < 1$, $0 < y < 1$ and $x + y > 1$ we have $f_{X,Y}(x,y) = 0$ but $f_X(x)f_Y(y) \neq 0$.)

6.4.3 Conditional probabilities and conditional densities

Example 6.11 continued. Suppose X and Y have joint density function

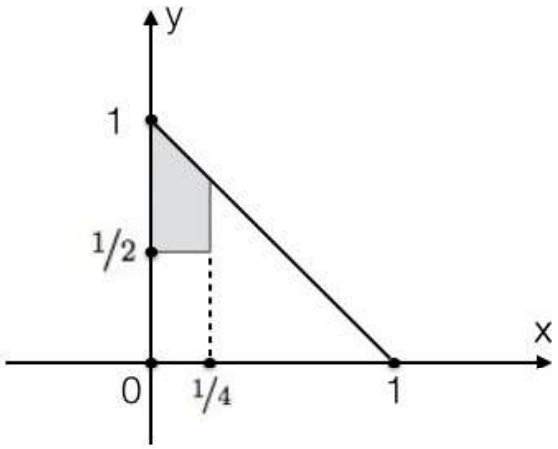
$$f_{X,Y}(x,y) = \begin{cases} 60xy^2 & x > 0,\ y > 0,\ x + y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Then from the definition of conditional probability for *events* we have

$$\mathbb{P}(Y \geqslant 1/2 \mid X \leqslant 1/4) = \frac{\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4)}{\mathbb{P}(X \leqslant 1/4)}$$

To compute the right hand side of the equation above, we note that:

- The numerator is the probability of a joint event, so we must use the joint density $f_{X,Y}$ to compute it.
- The denominator is the probability of a marginal event, so we can use the marginal density f_X to compute it.



$$\begin{aligned}
\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4) &= \int_{x=0}^{1/4} \left(\int_{y=1/2}^{1-x} f_{X,Y}(x,y)\ \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{1/4} \left(\int_{y=1/2}^{1-x} 60xy^2\ \mathrm{d}y \right) \mathrm{d}x \\
&= \int_{x=0}^{1/4} \left[20xy^3 \right]_{y=1/2}^{1-x} \mathrm{d}x \\
&= \int_{x=0}^{1/4} 20x(1-x)^3 - \frac{5x}{2}\ \mathrm{d}x \\
&= \int_{x=0}^{1/4} 20x(1-3x+3x^2-x^3) - \frac{5x}{2}\ \mathrm{d}x \\
&= \int_{x=0}^{1/4} \frac{35x}{2} - 60x^2 + 60x^3 - 20x^4\ \mathrm{d}x \\
&= \left[\frac{35x^2}{4} - 20x^3 + 15x^4 - 4x^5 \right]_{x=0}^{1/4} = \frac{35}{64} - \frac{5}{16} + \frac{15}{256} - \frac{1}{256} = \frac{37}{128}
\end{aligned}$$

$$\mathbb{P}(X \leqslant 1/4) = \int_0^{1/4} f_X(x)\ \mathrm{d}x = \int_0^{1/4} (20x - 60x^2 + 60x^3 - 20x^4)\ \mathrm{d}x = \frac{47}{128}\ (\textit{exercise: check this})$$

So:

$$\mathbb{P}(Y \geqslant 1/2 \mid X \leqslant 1/4) = \frac{\mathbb{P}(Y \geqslant 1/2,\ X \leqslant 1/4)}{\mathbb{P}(X \leqslant 1/4)} = \frac{37}{47} \approx 0.787$$

Question:
Suppose we have *observed* that $X = \frac{1}{4}$. Can we determine $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$ *even though* $\mathbb{P}(X = 1/4) = 0$?

Answer:
We can compute $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$ by using the conditional density for Y *given* $X = 1/4$.
We define the **conditional density** as follows:

Definition
Suppose X and Y are jointly continuous with joint density function $f_{X,Y}(x,y)$. Then the **conditional density of Y given $X = x$** is defined as

$$f_{Y|X}(y|x) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_X(x)} & \text{provided } f_X(x) > 0 \\ 0 & \text{otherwise} \end{cases}$$

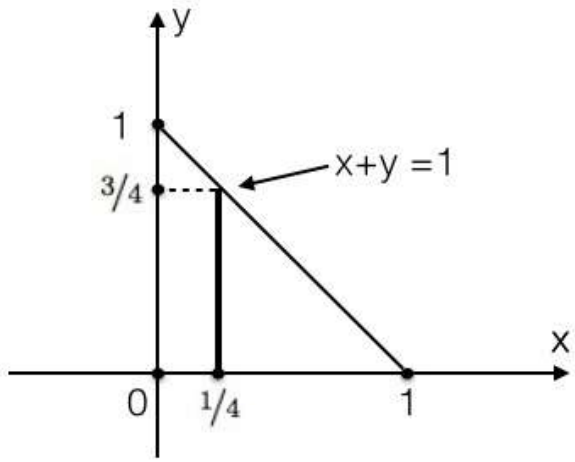
Likewise, the **conditional density of X given $Y = y$** is defined by

$$f_{X|Y}(x|y) = \begin{cases} \frac{f_{X,Y}(x,y)}{f_Y(y)} & \text{provided } f_Y(y) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Example 6.11 continued. We compute $\mathbb{P}(Y \geqslant 1/2 |\ X = 1/4)$ using the conditional density as follows:

Step 1: The first step is to determine the conditional density function $f_{Y|X}(y \mid 1/4)$. When working out the conditional density, we have to be careful to think about where the conditional

density $f_{Y|X}(y \mid 1/4)$ is non-zero. A good way to do this is to sketch the region where the joint density $f_{X,Y}(x, y)$ is non-zero and then draw the line corresponding to the *given information* that $X = 1/4$:



We can see from the diagram above that given $X = 1/4$ we must have $0 < Y < 3/4$, and that means that we must have $f_{Y|X}(y \mid 1/4) = 0$ for $y \leqslant 0$ or $y \geqslant 3/4$.

For $0 < y < 3/4$, the conditional density $f_{Y|X}(y \mid 1/4)$ is found as

$$f_{Y|X}(y \mid 1/4) = \frac{f_{X,Y}(1/4, y)}{f_X(1/4)} = \frac{60(1/4)y^2}{20(1/4) - 60(1/4)^2 + 60(1/4)^3 - 20(1/4)^4} = \frac{64}{9}y^2$$

That is,

$$f_{Y|X}(y \mid 1/4) \quad = \quad \begin{cases} 64y^2/9 & 0 < y < 3/4 \\ 0 & \text{otherwise} \end{cases}$$

Remark: It is easy to check that $f_{Y|X}(y \mid 1/4)$ is a valid density: it is clearly non-negative, and

$$\int_{y=0}^{3/4} f_{Y|X}(y \mid 1/4)\ \mathrm{d}y = \int_{y=0}^{3/4} \frac{64}{9}y^2\ \mathrm{d}y = \left[\frac{64y^3}{27}\right]_{y=0}^{3/4} = 1$$

Step 2: The next step is to use the conditional density $f_{Y|X}(y \mid 1/4)$ to compute the conditional probability $\mathbb{P}(Y \geqslant 1/2 \mid X = 1/4)$. Just as we can integrate density functions to compute unconditional probabilities, we can integrate *conditional* density functions to compute *conditional* probabilities:

$$\begin{aligned} \mathbb{P}(Y \geqslant 1/2 \mid X = 1/4) &= \int_{y=1/2}^{3/4} f_{Y|X}(y \mid 1/4)\ \mathrm{d}y \\ &= \int_{y=1/2}^{3/4} \frac{64y^2}{9}\ \mathrm{d}y = \left[\frac{64}{27}y^3\right]_{1/2}^{3/4} = 1 - \frac{8}{27} = \frac{19}{27} \approx 0.704 \end{aligned}$$

Note: The upper limit in the integral above is 3/4 because the conditional density $f_{Y|X}(y \mid 1/4)$ is only non-zero for $0 < y < 3/4$.

Remarks

1. In general, we can compute $\mathbb{P}(Y \in A \mid X = x)$ by integrating the conditional density $f_{Y|X}(y|x)$ over A, i.e.

$$\mathbb{P}(Y \in A \mid X = x) = \int_A f_{Y|X}(y|x)\ \mathrm{d}y$$

   Likewise,

$$\mathbb{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\ \mathrm{d}x$$

2. Suppose that the random variables X and Y are jointly continuous and independent. Then

$$f_{X|Y}(x|y) \overset{def}{\equiv} \frac{f_{X,Y}(x,y)}{f_Y(y)} \overset{indep}{=} \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x) \quad \text{(provided } f_Y(y) > 0\text{)}.$$

   It follows that if X and Y are independent,

$$\mathbb{P}(X \in A \mid\ Y = y) = \int_A f_{X|Y}(x|y)\ \mathrm{d}x = \int_A f_X(x)\ \mathrm{d}x = \mathbb{P}(X \in A)$$

   That is, if X and Y are independent, then *conditional probabilities* for X given information about Y are the same as *unconditional probabilities* for X. Similarly, for X, Y independent,

$f_{Y|X}(y|x) = f_Y(y)$ (provided $f_X(x) > 0$)
$\mathbb{P}(Y \in B \mid\ X = x) = \mathbb{P}(Y \in B)$.

3. 'Chain Rule' for densities. It follows from the definition given above that

$$\boxed{f_{X,Y}(x,y) = f_X(x)f_{Y|X}(y|x)}$$

   and

$$f_{X,Y}(x,y) = f_Y(y)f_{X|Y}(x|y)$$

   This is a *density* version of the Chain Rule. It means that we can compute the joint density from a *marginal density* and the appropriate *conditional density*

Recall the **Law of Total Probability (Partition Theorem)**: Suppose events $E_1, E_2, \ldots$ are disjoint with $\Omega = \cup_i E_i$, and $\mathbb{P}(E_i) > 0\ \forall i$. Then for *any* event F, we have

$$\mathbb{P}(F) = \sum_i \mathbb{P}(E_i)\mathbb{P}(F \mid E_i)$$

For density functions, a corresponding result is as follows.

Law of Total Probability (Partition Theorem) for densities
Suppose that X is a random variable with density f_X and suppose that B is an event. Then

$$\mathbb{P}(B) = \int_{\mathbb{R}} \mathbb{P}(B \mid X = x)f_X(x)\ \mathrm{d}x$$

Special Case
Suppose X and Y are jointly continuous random variables. Then

$$\mathbb{P}(Y \in A) = \int_{\mathbb{R}} \mathbb{P}(Y \in A \mid X = x) f_X(x)\ \mathrm{d}x$$

Proof.

$$\begin{aligned}
&\int_{\mathbb{R}} \mathbb{P}(Y \in A|\ X = x) f_X(x)\ \mathrm{d}x \\
&= \int_{\mathbb{R}} \left(\int_A f_{Y|X}(y|x)\ \mathrm{d}y \right) f_X(x)\ \mathrm{d}x \\
&= \int_A \left(\int_{\mathbb{R}} f_{Y|X}(y|x) f_X(x)\ \mathrm{d}x \right) \mathrm{d}y \qquad \text{(change order of integration)} \\
&= \int_A \left(\int_{\mathbb{R}} f_{X,Y}(x,y)\ \mathrm{d}x \right) \mathrm{d}y \qquad \text{(since } f_{X,Y}(x,y) = f_X(x) f_{Y|X}(y|x)\text{)} \\
&= \int_A f_Y(y)\ \mathrm{d}y \qquad \left(\text{since } \int_{\mathbb{R}} f_{X,Y}(x,y)\ \mathrm{d}x = f_Y(y) \right) \\
&= \mathbb{P}(Y \in A)
\end{aligned}$$

Example 6.13. Application of the Law of Total Probability. Let T denote the random time that Anne leaves her flat in the morning, measured in hours past 7 am, and let X denote the random time, measured in hours past 7am, when Anne's alarm clock rings. Suppose that the density for X is given by

$$f_X(x) = \begin{cases} 2x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

and suppose that, given $X = x$, the *conditional density* for T is given by

$$f_{T|X}(t|x) = \begin{cases} 2(t-x) & x < t < x+1 \\ 0 & \text{otherwise} \end{cases}$$

Note: We can see from the definition of the density f_X that $ImX = (0,1)$. That is, Anne's alarm clock rings at some time between 7am and 8am. Also, it follows from the definition of the conditional density $f_{T|X}(t|x)$ that *given* $X = x$, we must have $x < T < x+1$. That is, Anne leaves her flat at some time between the time when her alarm clock rings, and one hour after her alarm clock rings.

Now suppose that in order to catch her bus to the university, Anne must leave her flat before 8.30am. Determine the probability that tomorrow Anne will leave her flat before 8.30am, i.e. determine $P(T < 3/2)$.

To do this calculation, we can use the Law of Total Probability:.

$$\mathbb{P}(T < 3/2) = \int_0^1 \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x$$

Observe: If $x < 1/2$, then $\mathbb{P}(T < 3/2 \mid X = x) = 1$ because from the definition of the conditional density $f_{T|X}(t|x)$, we know that we always have $T < x + 1 < 3/2$ for $x < 1/2$.

If $1/2 \leq x < 1$, then

$$\begin{aligned}\mathbb{P}(T < 3/2 \mid X = x) &= \int_{t=x}^{3/2} f_{T|X}(t \mid x)\ dt \\ &= \int_{t=x}^{3/2} 2(t-x)\ dt \\ &= \left[(t-x)^2\right]_{t=x}^{3/2} \\ &= \left(\frac{3}{2} - x\right)^2\end{aligned}$$

So applying the Law of Total Probability we obtain:

$$\begin{aligned}\mathbb{P}(T < 3/2) &= \int_{x=0}^{1} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x \\ &= \int_{x=0}^{1/2} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x + \int_{x=1/2}^{1} \mathbb{P}(T < 3/2 \mid X = x)\ f_X(x)\ \mathrm{d}x \\ &= \int_{x=0}^{1/2} 1 \cdot 2x\ \ \mathrm{d}x \quad + \quad \int_{x=1/2}^{1} \left(\frac{3}{2} - x\right)^2 \cdot 2x\ \ \mathrm{d}x \\ &= \left[x^2\right]_{x=0}^{1/2} + \int_{x=1/2}^{1} \left(\frac{9}{4} - 3x + x^2\right) \cdot 2x\ \mathrm{d}x \\ &= \frac{1}{4} + \int_{x=1/2}^{1} \frac{9x}{2} - 6x^2 + 2x^3\ \mathrm{d}x = \frac{1}{4} + \left[\frac{9x^2}{4} - 2x^3 + \frac{x^4}{2}\right]_{x=1/2}^{1} \\ &= \frac{1}{4} + \left(\frac{9}{4} - 2 + \frac{1}{2}\right) - \left(\frac{9}{16} - \frac{1}{4} + \frac{1}{32}\right) = \frac{1}{4} + \frac{3}{4} - \frac{11}{32} = \frac{21}{32} \approx 0.656\end{aligned}$$

6.4.4 Conditional expectation

We can use the conditional density $f_{Y|X}(y|x)$ to compute the **conditional expectation of Y** given $X = x$ as follows:

$$\boxed{\mathbb{E}(Y \mid X = x) = \int_{\mathbb{R}} y\ f_{Y|X}(y|x)\ \mathrm{d}y}$$

Likewise, we can compute the **conditional expectation of X** given $Y = y$ by

$$\mathbb{E}(X \mid Y = y) = \int_{\mathbb{R}} x\ f_{X|Y}(x|y)\ \mathrm{d}x$$

Example 6.11 continued. For the joint density given in example 6.11, we showed previously that the conditional density $f_{Y|X}(y \mid 1/4)$ is given by $f_{Y|X}(y \mid 1/4) = 64y^2/9$ for $0 < y < 3/4$ (and zero elsewhere). Using this, we can compute

$$\begin{aligned}\mathbb{E}(Y \mid X = 1/4) = \int_{\mathbb{R}} y\ f_{Y|X}(y|1/4)\ \mathrm{d}y \ &= \ \int_0^{3/4} y \left(\frac{64y^2}{9}\right) \mathrm{d}y \\ &= \ \int_0^{3/4} \frac{64y^3}{9}\ \mathrm{d}y = \left[\frac{16}{9}y^4\right]_0^{3/4} = \frac{9}{16} = 0.5625\end{aligned}$$

Generalisation: More generally, if $g(Y)$ is a function of the random variable Y and $h(X)$ is a function of the random variable X, then we can use the conditional densities for X and Y to

compute:

$$\mathbb{E}\big(g(Y)|\ X=x\big)=\int_{\mathbb{R}} g(y)\ f_{Y|X}(y|x)\ \mathrm{d}y$$
$$\mathbb{E}\big(h(X)|\ Y=y\big)=\int_{\mathbb{R}} h(x)\ f_{X|Y}(x|y)\ \mathrm{d}x$$

Exercise: Show that when X and Y are jointly continuous and independent, then $\mathbb{E}(Y|\ X=x)=\mathbb{E}(Y)$ and $\mathbb{E}(X|\ Y=y)=\mathbb{E}(X)$.

Law of Total Expectation (Partition Theorem for Expectation)
Suppose that X and Y are jointly continuous with joint density function $f_{X,Y}(x,y)$. Then

$$\boxed{\mathbb{E}(X)=\int_{\mathbb{R}}\mathbb{E}(X\mid Y=y)f_Y(y)\,\mathrm{d}y}$$

Proof: Exercise.

Example 6.13 continued. Using the Law of Total Expectation, we can compute $\mathbb{E}(T)$, the expected time that Anne leaves her flat:

$$\mathbb{E}(T)\overset{\text{Law of Total Expectation}}{=}\int_{\mathbb{R}}\mathbb{E}(T\mid X=x)f_X(x)\,\mathrm{d}x=\int_0^1\mathbb{E}(T\mid X=x)\ 2x\ \mathrm{d}x \qquad (*)$$

Using the conditional density $f_{T|X}(t|x)=2(t-x)$ for $x<t<x+1$, we compute

$$\begin{aligned}\mathbb{E}(T\mid X=x)=\int_{t=x}^{x+1} t f_{T|X}(t|x)\ dt &= \int_{t=x}^{x+1} 2t(t-x)\ dt\\ &= \int_{t=x}^{x+1} 2t^2-2tx\ dt\\ &= \left[\frac{2t^3}{3}-t^2x\right]_{t=x}^{x+1}\\ &= \left(\frac{2(x+1)^3}{3}-(x+1)^2x\right)-\left(\frac{2x^3}{3}-x^3\right)\\ &= x+\frac{2}{3}\end{aligned}$$

and substituting this into $(*)$ yields

$$\mathbb{E}(T)=\int_0^1\left(x+\frac{2}{3}\right)2x\ \mathrm{d}x=\int_0^1 2x^2+\frac{4x}{3}\ \mathrm{d}x=\left[\frac{2x^3}{3}+\frac{2x^2}{3}\right]_0^1=\frac{4}{3}\text{ hours}$$

Recall that T is measured in hours past 7am, so the expected time that Anne leaves her flat is 8:20am.